

DesignWorkBench v2

Constructing a Professional Design Benchmark for Coding Agents

Sarvesh Chidambaram

Memi Design, United States

ORCID: not supplied | Correspondence through [memi-design/memi](mailto:memi-design@memi.com)

Research artifact version 1.0 | 29 July 2026 | Benchmark schema 2

Abstract

Benchmarks for coding agents commonly score tests, screenshots, or short interactions. Professional design work additionally requires evidence judgment, editable source, platform fidelity, accessibility, critique, validation, and a handoff another practitioner can continue. We introduce the construction protocol for DesignWorkBench v2, a multidisciplinary benchmark that treats design as complete work rather than visual imitation. The frozen foundation specifies 300 task contracts across 15 tracks, partitioned into 60 public development tasks, 180 private tests, and 60 rolling holdouts. Tasks span 30–165 minutes and bind expected artifacts, runtime runners, provenance, negative controls, and discipline-weighted rubrics. Functional acceptance, senior-practitioner quality, source trust, and efficiency remain separate outputs. Calibration requires at least 4 qualified practitioners per track, 2 external to the benchmark author, blinded ratings, and chance-corrected agreement. The current artifact is deliberately incomplete: 60 public fixtures are prepared but 0 are independently verified, 3 of 8 runner profiles have reproducible evidence, and no practitioner calibration has been collected. We also preregister a matched baseline, Memi, Superpowers, and Everything Claude Code comparison without inventing results. This paper contributes a falsifiable benchmark design and readiness accounting, not a claim that Memi has reached senior-practitioner performance.

Keywords: design benchmark; coding agents; professional practice; practitioner calibration; artifact provenance; human evaluation

1 Motivation

Screenshot similarity and unit tests capture useful but narrow properties of an interface. They do not establish that an agent framed the right problem, used trustworthy evidence, preserved an editable system, handled error and recovery states, respected platform conventions, or left a usable handoff. Design2Code combines automated and human evaluation for screenshot-to-code generation [11]; WebArena emphasizes reproducible, long-horizon interaction [13]; and BigLaw Bench shows the value of practitioner-authored tasks, bespoke rubrics, negative criteria, and separate work-product and source-trust scores [4].

Benchmark construction itself is a failure surface. SWE-bench Verified was created after expert review removed tasks that were hard or impossible to solve [9]; a later audit reported test defects and contamination that reduced its usefulness for frontier evaluation [10]. DesignWorkBench therefore treats fixture verification, leakage controls, runner evidence, and benchmark evolution as first-class research objects.

The benchmark asks:

1. Can an agent complete an end-to-end professional assignment?
2. Is the artifact functionally accepted in the required runtime?
3. How close is the work to a qualified senior practitioner’s deliverable?
4. Are claims traceable to supplied evidence and editable source?
5. When quality is non-inferior, what tokens, time, retries, tools, and human intervention were required?

2 Construction Principles

2.1 Separate outcomes

The benchmark reports four non-substitutable outcomes:

Acceptance.

Deterministic build, runtime, interaction, integrity, and artifact checks.

Professional quality.

Blinded, rubric-based judgment calibrated against qualified working practitioners.

Source trust.

Traceability from consequential claims to supplied sources, decisions, and evidence.

Efficiency.

Tokens, latency, tools, retries, interventions, and verification coverage, evaluated only after quality non-inferiority.

The separation blocks three common shortcuts: treating a passing build as expert design, treating a polished screenshot as working software, and claiming cost savings after silently lowering quality.

2.2 Evidence caps

Scores are bounded by the strongest admitted evidence. Prompt-only claims are capped at 25; implemented and tested work at 65; runtime-reproduced and independently validated evidence may reach 100. A model cannot award itself practitioner status. Missing evidence is recorded as missing rather than imputed from another track.

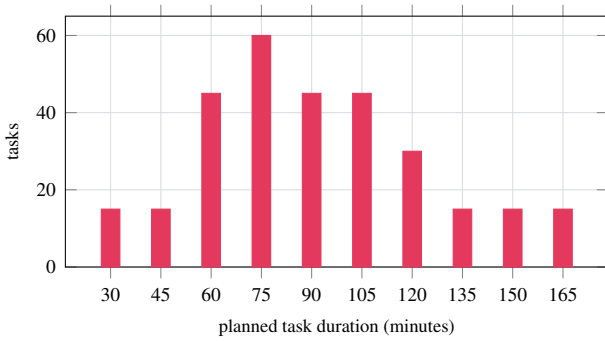


Figure 1: Task-duration distribution. Longer contracts test system changes, cross-functional validation, and recovery rather than single-screen output.

2.3 Frozen units and immutable receipts

Each evaluation cell binds a task revision, fixture hash, repository revision, runtime profile, model, provider, reasoning setting, condition revision, artifact bundle, and trace receipt. Generated candidates are not promoted by editing status fields. A separate reviewer recomputes the candidate hash and issues an immutable receipt.

3 Task Bank

3.1 Tracks and modes

The task bank contains 15 professional tracks: product; graphic and brand; motion; spatial, AR, and VR; agentic; iOS and SwiftUI; cross-platform mobile and Expo; Android and Compose; native design kits; design systems; design engineering; service; content; interaction and prototyping; and research and evidence.

Each track contains 20 contracts and five recurring work modes: critique and diagnose, create, modify in context, validate and hand off, and capstone recovery. The contracts range from 30 to 165 (90-minute median) and require both discipline artifacts and a decision log, execution trace, and handoff note.

3.2 Splits

The 300 contracts are frozen as 60 public development, 180 private test, and 60 rolling holdout tasks. Public tasks support implementation and reproducibility. Private tasks measure generalization without exposing exact fixtures. Holdouts rotate quarterly to limit prompt memorization and benchmark-specific tuning. Practice or public work cannot be relabeled as private evidence.

3.3 Artifact contracts

Every task declares required input kinds, editable output kinds, runner profiles, acceptance fixtures, and negative controls. Required artifacts are not interchangeable. For example, a native task cannot be satisfied by a web mock, a layered design task cannot be satisfied by a flattened image, and a service-design task cannot omit offline operations and support channels.

4 Rubrics and Negative Controls

4.1 Professional dimensions

Track-specific weights sum to 100 across ten shared dimensions: problem framing; evidence judgment; craft; interaction behavior; system and platform fidelity; accessibility, safety, and ethics; technical runtime quality; validation; handoff and collaboration; and critique and recovery. Shared dimensions support comparison, while track weights preserve professional differences. Native-platform references include Apple and Android quality guidance [1, 2]; web artifacts incorporate WCAG 2.2 and field-oriented performance criteria [3, 12]; spatial tasks bind OpenXR-compatible evidence [6].

The benchmark uses a geometric composite across claimed tracks. A high web score cannot numerically conceal a failed native, spatial, motion, or source trust result. Every track is also reported independently.

4.2 Negative controls

The manifest defines 18 named failure controls. They include fixture tampering, fabricated evidence, hidden-test leakage, scope substitution, unsupported user needs, flattened artifacts, unsafe motion, spatial comfort failures, silent agent overwrites, non-native substitution, single-platform proof, duplicate-component debt, polished front ends without required state, placeholder content, happy-path-only prototypes, and unsupported research claims.

Negative controls are not optional deductions added after a favorable score. They can invalidate evidence, cap scores, or fail acceptance. Their role is to represent the human correction cost of plausible-looking but unsafe or unusable work.

5 Runner and Fixture Evidence

5.1 Runner contracts

Eight runner profiles define evidence for browser and desktop web, iOS or visionOS Simulator, Expo on iOS and Android, Android emulator and Compose, editable design artifacts, motion rendering, spatial/XR, and cross-discipline artifact provenance. A runner is verified only when every evidence kind in its contract is reproducible and hash-bound.

At this paper revision, browser/Playwright, motion rendering, and artifact validation are **VERIFIED**. The remaining five are **PENDING**. Tool installation, capability detection, or a screenshot alone does not verify a runner.

5.2 Fixture promotion

The public authoring pass produced 60 deterministic candidate packs. All 0 are independently verified. Promotion requires: reproducible source assets; license or permission evidence; private-material separation; matching hashes; complete artifact and runtime contracts; and a clean-environment handoff reopen. Benchmark authors cannot self-approve their own candidates.



Figure 2: Predeclared cross-harness comparison. Conditions cannot select different tasks, models, acceptance contracts, or graders.

6 Practitioner Calibration

Each track requires at least 4 qualified practitioners, including 2 external to Memi. Qualification and conflicts are recorded per track. Each practitioner completes at least 5 frozen artifacts, and every artifact receives at least 2 independent blinded ratings. Graders do not receive the author, model, provider, harness condition, another rating, or aggregate result.

Ratings contain dimension scores, rationale, acceptance, evidence level, artifact hash, qualification reference, and start and completion times. Disagreements above eight points require a third adjudicator while preserving the original ratings. Interval-scale Krippendorff’s alpha [5] must reach 0.8 overall and at least 0.67 in every claimed track. Current calibration contains 0 practitioners and 0/15 calibrated tracks; professional certification is therefore **BLOCKED**.

7 Cross-Harness Evaluation Protocol

The committed protocol defines 4 conditions: no repository design harness, Memi 2.7, Superpowers, and Everything Claude Code. The Skills CLI is classified as installation and discovery transport, not as a performance condition. The protocol is *preregistered and unexecuted*.

The experimental unit is task–repository–model–repeat. Repository and task revisions, fixture hash, model, provider, reasoning effort, context window, permissions, network policy, runtime image, time limit, acceptance contract, and grader blinding are held fixed. Condition order uses a balanced Latin square. Development cells require 3 independent runs; release-critical holdouts require 5.

Primary outcomes are acceptance, senior work-product score, source trust, and critical defect count. Secondary outcomes are total and uncached tokens, wall time, tool calls, retries, human interventions, and verification coverage. Analysis is paired and stratified by track, with paired bootstrap intervals and exact sign tests. A harness is ranked only after all admitted conditions complete the same frozen cells and quality non-inferiority passes. Structured authentication or setup failures are reported by cause, not converted into performance zeros or

Table 1: Current cross-harness result status. Dashes are absent measurements, not zero scores.

Condition	Admitted runs	Quality	Efficiency
No design harness	0	–	–
Memi 2.7	0	–	–
Superpowers	0	–	–
Everything Claude Code	0	–	–

Table 2: Foundation and certification are deliberately separate.

Evidence unit	Prepared	Verified	Status
Task contracts	300	300	frozen
Public fixture packs	60	0	BLOCKED
Runner profiles	3	3	partial
Practitioners	0	0	BLOCKED
Calibrated tracks	0	0	BLOCKED
Cross-harness cells	0	0	BLOCKED

wins for another condition.

8 Current Readiness

The benchmark foundation validates structurally: 15 tracks, 300 contracts, expected split counts, rubric totals, task hashes, and runner contracts. This is not a quality score. It shows that the measurement instrument is defined and machine-checkable. Certification remains blocked on external work.

9 Threats and Governance

Author involvement. Memi authors designed the benchmark and may benefit from its conclusions. External fixture review, blinded grading, immutable negative evidence, and public protocols reduce but do not remove this conflict.

Construct validity. Ten dimensions cannot exhaust every design practice. Track rubrics and practitioner rationales must be reviewed after calibration without retroactively changing frozen evaluations.

Task realism. The 60 public candidate packs are synthetic development material. They are useful for deterministic tooling but cannot establish ecological validity until independently verified and complemented by private and holdout tasks.

Provider and harness drift. Models, harnesses, and dependencies change. Every run pins versions, and comparisons may not mix revisions without reporting them as separate conditions.

Contamination. Public tasks are assumed learnable. Private fixtures are access-controlled, holdouts rotate, and leaked tasks are retired rather than silently reused.

Metric gaming. Optimizing a geometric composite can still produce artifacts that exploit rubric gaps. Negative controls, free-text practitioner rationales, runtime evidence, and periodic adversarial audits are required.

Benchmark releases use semantic versions. Task edits change hashes; rubric or split changes require a benchmark-version change; and results always identify the benchmark, candidate, runner, and condition revisions.

10 Reproducibility and Availability

The benchmark manifest, public candidate fixtures, validation scripts, runner receipts, calibration templates, preregistered cross-harness protocol, paper source, generated tables, and readiness report are public [7, 8]. Private and holdout fixtures are not published. Public evidence retains content hashes so disclosed artifacts can be verified without exposing protected source.

```
npm run build:designwork-bench
npm run check:designwork-bench
npm run build:designwork-evidence
npm run check:designwork-evidence
npm run build:designwork-readiness
npm run check:designwork-readiness
npm run check:designwork-release
```

The ORCID field remains “not supplied” until the author provides a verified identifier. This PDF is a research artifact and has not been peer reviewed.

11 Conclusion

DesignWorkBench v2 defines a benchmark for complete professional work: discipline-specific artifacts, live runtimes, evidence trust, negative controls, senior calibration, critique, recovery, and handoff. Its main contribution is an evidence boundary that can refuse flattering conclusions. The task bank and three runner profiles are reproducible; independent fixtures, practitioner reliability, provider diversity, and matched cross-harness runs remain incomplete. Those blocked cells are the next work, not numbers to infer.

References

- [1] Apple. Human interface guidelines. <https://developer.apple.com/design/human-interface-guidelines>, 2026. Accessed 29 July 2026.
- [2] Google. Core app quality. <https://developer.android.com/develop/adaptive-apps/quality-guidelines/core-app-quality>, 2026. Accessed 29 July 2026.
- [3] Google. Web vitals. <https://web.dev/articles/vitals>, 2026. Accessed 29 July 2026.
- [4] Harvey. Introducing biglaw bench. <https://www.harvey.ai/blog/introducing-biglaw-bench>, 2024. Accessed 29 July 2026.
- [5] John Hughes. krippendorffsalph: An r package for measuring agreement using krippendorff’s alpha coefficient. *The R Journal*, 2021. URL <https://journal.r-project.org/articles/RJ-2021-046/>.
- [6] Khronos Group. Openxr conformance test suite usage. https://registry.khronos.org/OpenXR/conformance/cts_usage.html, 2026. Accessed 29 July 2026.
- [7] Memi Design. Memi designworkbench v2 benchmark manifest. <https://github.com/memi-design/memi/blob/main/benchmarks/designworkbench-v2/benchmark.json>, 2026.
- [8] Memi Design. Designworkbench v2 readiness report. <https://github.com/memi-design/memi/blob/main/docs/audits/memi-designworkbench-v2-readiness.md>, 2026.
- [9] OpenAI. Introducing swe-bench verified. <https://openai.com/index/introducing-swe-bench-verified/>, 2024. Updated 24 February 2025; accessed 29 July 2026.
- [10] OpenAI. Why swe-bench verified no longer measures frontier coding capabilities. <https://openai.com/index/why-we-no-longer-evaluate-swe-bench-verified/>, 2026. Accessed 29 July 2026.
- [11] Chenglei Si, Yanzhe Zhang, Ryan Li, Zhengyuan Yang, Ruibo Liu, and Diyi Yang. Design2code: Benchmarking multimodal code generation for automated front-end engineering. In *Proceedings of NAACL*, 2025. URL <https://aclanthology.org/2025.naacl-long.199/>.
- [12] World Wide Web Consortium. Web content accessibility guidelines (wcag) 2.2. <https://www.w3.org/TR/WCAG22/>, 2024.
- [13] Shuyan Zhou, Frank F. Xu, Hao Zhu, et al. Webarena: A realistic web environment for building autonomous agents. *arXiv preprint arXiv:2307.13854*, 2023. URL <https://arxiv.org/abs/2307.13854>.

A Runner Evidence Contracts

Runner	Current evidence	Required evidence contract
Browser and desktop web runtime	verified	browser-build, playwright-e2e, accessibility-tree, performance-trace
iOS or visionOS simulator runtime	pending	xcode-build, simulator-capture, accessibility-capture, performance-trace
Expo development and production runtime	pending	expo-build, ios-capture, android-capture, navigation-e2e
Android emulator and Compose instrumentation	pending	gradle-build, compose-ui-test, emulator-capture, accessibility-capture
Editable design artifact runtime	pending	layer-tree, component-graph, token-bindings, editable-export
Frame-accurate motion and video runtime	verified	timeline-source, render, frame-analysis, reduced-motion-output
Spatial simulator or OpenXR-compatible runtime	pending	scene-build, interactive-scenario, input-profile, comfort-safety-check
Cross-discipline artifact and provenance validator	verified	schema-validation, hash-verification, source-provenance, handoff-reopen

B Evidence Commitments

Artifact	SHA-256
benchmarks/designworkbench-v2/benchmark.json	69ebcc306e4751452e7dad12850f5be2b43d5186a9b3aa66e683339638cff9e9
docs/audits/memi-designworkbench-v2-readiness.json	2c6cdfde67405664d32e4838ee5146de023e6e57df0abc77a9ddc1dc31e2939e
benchmarks/designworkbench-v2/PRACTITIONER_CALIBRATION.md	969f1853baf1006d24df961628ee33f695665e009f8949e6216d8824bda15f82
benchmarks/designworkbench-v2/evidence/evidence.json	8c9fc9817f823827a76362bb6fee3aa2e5bed152d05d856ec6628ace7f99e012
docs/research/designworkbench-v2-paper/cross-harness-protocol.json	9a5a9d51d58a0ad27db080bb08b167e5f92191006d64a2b02b3705eedc3cf22c

C Track Inventory

Code	Track	Required discipline artifact families
T1	Product design	research synthesis, journey, IA, prototype, metrics
T2	Graphic and brand	concept, identity system, compositions, production files
T3	Motion	storyboard, timing system, animation source, render, reduced motion
T4	Spatial/XR	spatial map, scene source, interaction model, safety evidence
T5	Agentic design	authority model, state/trace UI, conflict and recovery flow
T6	iOS/SwiftUI	native source, Simulator flow, accessibility, performance trace
T7	Cross-platform mobile	Expo source, iOS/Android captures, navigation E2E
T8	Android/Compose	Compose source, emulator flow, UI tests, accessibility
T9	Native design kits	tokens, native components, platform mappings, usage guide
T10	Design systems	tokens, components, governance, migration, adoption evidence
T11	Design engineering	production source, states, tests, runtime, code handoff
T12	Service design	research, blueprint, channel artifacts, operating model
T13	Content design	evidence, content model, state copy, accessibility, governance
T14	Interaction/prototyping	flows, state model, interactive source, usability evidence
T15	Research/evidence	plan, corpus, analysis, findings, decision trace